

1. Base Case Proof:

- When $n=0$, $\text{Square}(0)$ returns $0^2 = 0$. This proves the base case that $\text{Square}(n)$ returns n^2 for $n \geq 0$.

2. Strong Induction Hypothesis:

- Assume that for all k where $0 \leq k \leq n$, $\text{Square}(k)$ return k^2 . This is the strong induction hypothesis.

3. Induction Goal:

- The goal is to prove that $\text{Square}(n)$ returns n^2 for $n \geq 0$, assuming the strong induction hypothesis.

4. Induction Step Proof:

- We need to consider two cases:

Case 1 : n is even

If n is even, i.e. $n=2y$ for some integer y .

- Using the strong induction hypothesis, let $t = \text{Square}(y)$.
- Since t is square of y and $4t$ is the square of $2y$.
- Therefore, $\text{Square}(n) = 4t = 4y^2 = n^2$.

Case 2 : n is odd

If n is odd, i.e. $n=2y+1$ for some integer y .

- Using the strong induction hypothesis, let $t = \text{Square}(y)$.
- t is the square of y , and $4t + 2y + 1$ is the square of $2y+1$.

· Therefore, $\text{Square}(n) = 4t + 2n - 1 = 4y^2 + 2(2y + 1) - 1 = n^2$

This completes the induction step, proving that if the statement holds for all k up to n , then it holds for n .